

13 • Project Plan, Presentation & Contributions

Assignment: "Include a final slide describing individual contributions." • "Duration: 10 minutes for presentation, 10 minutes for questions."

13.1 Status

Assignment item	Artefact	Status
Health / health system need	01	✓
User personas & scenarios	02	✓ 3 personas · 2 required scenarios + 1 stress case
Guidelines and evidence (L1)	03	✓ 4 sources, appraised
L2 — high-level workflow (BPMN)	04 + diagrams/	✓ 2 diagrams, typed tasks, English
L2 — data dictionary	05	✓ 48 elements
L2 — decision logic	06	✓ 18 rules
L2 — L1→L2 challenges	07	✓ 9 documented
L3 — machine-readable code	08 + l3/	✓ v0.3.4, bilingual, FHIR configured, graphics embedded
L4 — executable layer	09 + l4/	✓ single file, no build step, bilingual
QA — risk-based testing	10	✓ 19/19 logic · 20/20 FHIR · no-hardcoding check · 8 defects fixed
Implementation, M&E strategy	11	✓
Expected impact	12	✓ incl. SaMD classification
Presentation	submission/	✓ 16 slides + 10 backup slides, with speaker notes
Presentation script	14	✓ word for word, with timings, handovers and Q&A jump targets
Demo	live click-through of the L4 on slide 11	✓ choreography in 14 §SLIDE 11 · submission/WundCheck_L4_Demo.mp4 (1:46) kept as the emergency fallback
Submission package	submission/	✓ 16 PDFs + L3 + L4 + BPMN sources + demo video + index

13.2 Remaining Work

Three items, none of which we can close without the group:

#	Task	Why it cannot be automated	Time
1	Set up the live demo on the presentation laptop	The click-through must be rehearsed on the actual machine — see the checklist in 14 §14.5	20 min

#	Task	Why it cannot be automated	Time
2	Rehearse twice with a stopwatch	Following 14; the first run will overrun — cut in the given order, do not speed up	40 min
3	Clarify the submission channel	Ask the lecturers during Tuesday morning's practical	5 min

Optional, if time allows: redraw the wound graphics against the Southampton grades and for more than one skin tone (the current set is schematic and shows a single skin tone — this is named as residual risk (b) and (d) in 10); add a graphic for `redness_extent`.

13.3 Presentation — 16 Slides, two speakers

Full spoken text for every slide, including the backups: **14 • Presentation script.**

#	Slide	Speaker	Key message	Time
1	Title	Feline	WundCheck — wound check in the hands of the patient	0:15
2	Health need & scope	Feline	SSIs occur after discharge — exactly when nobody is looking	0:42
3	Personas & Scenario A	Feline	The call that is avoided is worth as much as the one that is triggered	0:35
4	L1 & evidence appraisal	Feline	We discarded the statistically better instrument — it cannot be captured by laypersons	0:42
5	Southampton mapping	Feline	Our choice of images is a published grading, not a UI whim	0:42
6	BPMN high-level	Feline	Four lanes, one business rule task — the logic is not in the app	0:30
7	Data dictionary	Feline	48 elements · 9 derived · 38 with terminology · 47 FHIR-mapped	0:25
8	Decision table & worst-first	Feline	False reassurance weighs more than a false alarm	0:38
9	L1→L2 challenges	Feline	Nine places where the guideline left us on our own	0:38
10	L3 architecture decision	Sivanajani	Author custom JSON, speak standards at the interface — and <i>prove</i> the app holds no clinical content	0:30
11	Live demo	Sivanajani	Green · orange + FHIR · changing a threshold without a rebuild	1:50
12	QA: hazards, 8 defects, residual risk	Sivanajani	Three defects were invisible to document review	0:45
13	Implementation & M&E	Sivanajani	What we could not test, we monitor in deployment	0:35
14	Impact & SaMD	Sivanajani	IMDRF Category II — with a rationale	0:28
15	Individual contributions	Sivanajani	Testing was cross-assigned; neither signed off their own layer	0:15
16	Repository & QR code	Sivanajani	Everything is reproducible — scan and re-run the checks	0:12

Total: 9:42, inside the 10-minute limit. 14 §14.1 lists, in order, what to cut if the talk runs long.

Backup slides 17–26 (*not presented — for the Q&A*)

17 complete decision table · 18 test protocol T1–T19 · 19 hazard analysis and risk matrix · 20 BPMN detail process · 21 FHIR interface · 22 traceability L1 → FHIR · 23 the L3 in one screen · 24 data dictionary columns · 25 depth that did not fit into ten minutes · 26 sources

Each has its own 20–30 s script in 14 §14.3; jump targets per likely question are tabulated in 14 §14.4.

The key figures line for slide 7

48 data elements • 9 derived • 38 with terminology binding • 47 with FHIR mapping • 18 decision rules • 9 documented L1→L2 challenges • 6 hazards • 39 automated checks (19 decision logic + 20 FHIR conformance) plus a machine-enforced no-hardcoding check, all passing • 8 defects found and fixed

13.4 The Narrative Thread

“We translated a guideline written for health professionals into an instrument for laypersons — and in doing so learned where that translation becomes dangerous.”

This line carries all 10 minutes:

- **L1** — CDC and Southampton are written for clinicians who examine the wound. Campwala et al. show that no instrument is perfect, and that the best one is unusable for us
- **L2** — Every rule had to be rebuilt so that a 58-year-old patient can answer it at home. Eight of these translations are documented decisions that come at a price
- **L3/L4** — The choice of images is the *answer* to the translation problem, not a UI gimmick. And the logic stays in the L3 so that clinical staff can maintain it without developers
- **QA** — The errors lay exactly where translation had taken place: conditional logic made an emergency rule unreachable, a captured symptom fed into nothing
- **Impact** — A triage tool in the hands of patients is regulatorily delicate. False reassurance weighs more heavily than a false alarm. Hence worst-first, hence the residual risk statement, hence the monitoring indicator “GREEN despite presentation to a physician”

13.5 Q&A Preparation

Both of us must be able to explain the entire project — the question round is addressed to everyone and accounts for 20 % of the grade. Backup-slide jump targets per question: 14 §14.4.

Question	Where the answer is
Why these guidelines and not others?	03 §3.2
Why Southampton, when the study attests only limited value to it?	03 §3.3 — triage vs. prognosis
Why not ASEPSIS, which performed better?	07 C-06 — cannot be captured by laypersons
What was your most difficult translation decision?	07 C-01 and C-03
Where is your logic most fragile?	Image assignment by laypersons — residual risk (b), 10 §10.5
Why no XLSForm?	08 §8.1 — comparison table
How do you roll out a guideline update? How long does that take?	08 §8.6 and 11 §11.4 — target ≤ 4 weeks
What did you not test, and why?	10 §10.5 — residual risk statement

Question	Where the answer is
Is this a medical device? Which category?	12 §12.4 — SaMD, IMDRF II, with a rationale
How do you prevent automation bias?	12 §12.3 — the result always states the reason
What happens with a case that fits no rule?	Catch-all R8 — never phrases things as “everything is fine” without a note to make contact
How do you connect to a HIS?	11 §11.2 — FHIR R4
Which interoperability level do you achieve?	09 §9.4 — structural yes, semantic partially, organisational no
What would be your primary outcome in an evaluation?	11 §11.8 — time to medical contact
Who is liable for an incorrect recommendation?	12 §12.3 — an open question, to be named honestly
What would you do differently given more time?	12 §12.5

13.6 Individual Contributions

Person	Area of responsibility	Artefacts	Presents
Feline Weger · Biomedical Engineer	Clinical lead & L2 — L1 source research and evidence appraisal, health need and scope, personas and scenarios, Southampton → image mapping, the nine L1→L2 translation decisions, BPMN, data dictionary, decision table, terminology mapping	01, 02, 03, 04, 05, 06, 07, diagrams/	Slides 2–9
Sivanajani Sivakumar · MSc Medical Informatics	Technical lead & QA — L3 schema and authoring, the L4 application and expression engine, FHIR R4 interface, hazard analysis, test protocol, no-hardcoding check, bug log, implementation and M&E strategy, impact and regulatory classification	08, 09, 10, 11, 12, l3/ , l4/ , tools/	Slides 10–16

Testing was cross-assigned on purpose: neither of us signed off the layer we wrote. Both of us can answer questions on the whole project — the question round is addressed to everyone and accounts for 20 % of the grade.

13.7 Submission Package

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submission/
├─ 00_SUBMISSION_INDEX.md / .pdf
├─ WundCheck_CDSS_Presentation.pptx / .pdf    (15 slides + 10 backup slides, speaker
notes)
├─ WundCheck_L4_Demo.mp4                      (1:46, screen recording of the running app)
├─ L3_wundcheck-l3.json                      (the machine-readable L3)
├─ L4_wundcheck-app.html                     (single file, offline, EN/DE)
├─ BPMN_workflow-highlevel.bpmn / .svg / .png
├─ BPMN_workflow-app-detail.bpmn / .svg / .png
├─ example-fhir-bundle.json
├─ README.pdf
└─ 01 ... 14 – every document as PDF

```

All documents additionally as PDF — do not rely on the lecturers having a Markdown viewer or bpmn.io at hand. The repository itself is the long version, the PDF package is the submission.

- ☐ **Clarify the submission route** — ask directly in the practical session on Tuesday morning
- ☐ Demo video stored **locally** on the presentation laptop, not only in the cloud

- ☐ Backup: screenshots of the demo in the slides in case the video does not start